

Performance Testing

Date	15 July 2026
Team ID	XXXXXX
Project Name	XXXXXX
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	Loan Prediction API (/predict)
Test Environment	Local Machine (Flask Application)
Test Date	15 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	1	Predict loan approval with valid applicant data	10	60 sec
2	2	Multiple users submitting loan applications simultaneously	50	120 sec
3	3	Continuous prediction requests	100	180 sec

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 sec	1.28 sec	Pass	Fast response
2	Response Time (Max)	< 5 sec	3.72 sec	Pass	Within acceptable limit
3	Throughput	-	72 Requests/sec	Pass	Stable throughput
4	Error Rate	<1%	0.3%	Pass	No significant failures
5	CPU Utilization	<80%	58%	Pass	Efficient resource usage
6	Memory Utilization	<80%	61%	Pass	Memory usage remained stable

Step 4: Observations & Analysis

Key Findings

- The Smart-Lender application successfully handled concurrent loan prediction requests.
- Average response time remained below the target value.
- No significant performance degradation was observed during normal load.

Bottlenecks Identified

- Response time increased slightly when more than 150 virtual users accessed the system simultaneously.
- Model prediction took slightly longer under peak load.

Optimization Steps Taken

- Optimized Flask API request handling.
- Improved preprocessing of input data.
- Reduced unnecessary computations before prediction.
- Enabled model reuse instead of loading the model for every request

